



TAXONOMY — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

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1. Taxonomic Hierarchy & Species

- Plant taxonomy ka hierarchy higher se lower: Kingdom → Phylum → Class → Order → Family → Genus → Species. Fundamental similarities wale individual organisms ka group species hota hai; related species milkar genus banate hain. [Q1, Q2]
- Source ke mutabik identified species ki sankhya mein insects sabse bada group hain; Insecta, Arthropoda phylum ka class hai. [Q3]

2. Bacteria, Microbes & Unicellular Reproduction

- Antonie van Leeuwenhoek ko source bacteria ki discovery aur father of microbiology se associate karta hai. Bacteria microscopic hote hain aur compound microscope se dekhe ja sakte hain. [Q4, Q5]
- Kuch bacteria autotrophic hote hain; photosynthetic bacteria mein plant-like chloroplast nahi, bacteriochlorophyll pigments hote hain. Bacteria living, single-celled prokaryotic microbes hain; living–non-living borderline wali baat viruses ke liye di gayi hai. [Q6, Q7]
- Gram-positive vs Gram-negative distinction cell-wall structure par based hai. Mycoplasma source ke mutabik autonomous growth/reproduction capable smallest organism hai aur isme cell wall absent hoti hai. [Q8, Q9]
- Amoeba binary fission se reproduce karta hai; source reason “all unicellular organisms reproduce asexually” ko false batata hai because some unicellular eukaryotes sexual reproduction bhi kar sakte hain. [Q10]
- Bacteria aur fungi artificial/synthetic medium par cultured ho sakte hain, lekin viruses ko living host cell chahiye. [Q12]

EXAM TRAP — Source has multi-correct/disputed items

- ▶ Q11 mein spore formation Fungi, Algae aur Fern—multiple listed organisms mein possible diya gaya hai; source ise disputed / more than one option correct note karta hai. [Q11]
- ▶ Q12 ka clear distinction: bacteria + fungi artificial medium par; virus living host ke bina replicate nahi karta. [Q12]

3. Animal Physiology — Warm-blooded, Respiration & Basic Classification

- Warm-blooded mammals aur birds external environment ke bawajood body temperature relatively constant rakhte hain; source ise homeostasis se relate karta hai. [Q13]
- Echinoderms generally external fertilization aur egg-laying pattern dikhate hain; source Q14 mein “echinoderms are viviparous” ko incorrect statement batata hai. [Q14]
- Hydra mein blood/circulatory system nahi hota, phir bhi respiration aur excretion cell layers ke across simple diffusion se hoti hai. [Q15]

4. True Fish, Birds & Marine Mammals

- Starfish true fish nahi; source ise Echinodermata mein rakhta hai. Sawfish, pipefish aur guitarfish true fishes hain. Catfish aur seahorse bhi true fishes diye gaye hain; silverfish insect, crayfish crustacean, jellyfish cnidarian aur sea cucumber echinoderm hai. [Q16, Q17, Q18, Q19]
- Shark ka skeleton bone ka nahi, cartilage/connective tissue ka hota hai; source answer “0 bones” deta hai. Golden Mahseer fish hai, bird nahi. [Q20, Q21]
- Blue whale largest mammal hai; whales mammals hain. Sperm whale largest toothed whale hai, while blue whale baleen whale hai. [Q22, Q23, Q24, Q25]
- Echidna egg-laying mammal hai; bats true sustained flight capable mammals hain aur Aves group ke member nahi. [Q26, Q27, Q28, Q29]
- Sea cow/dugong aur sea lion mammals hain, seahorse fish hai. Dolphins aquatic mammals / toothed whales ke family group mein diye gaye hain; seals pinniped marine mammals hain. [Q31, Q32, Q33]

EXAM TRAP — Indian Flying Fox (source-specific current note)

- ▶ Source explanation ke mutabik Indian Flying Fox pehle Wildlife Protection Act, 1972 ke Schedule V/vermin category mein tha, lekin 2022 amendment ke baad Schedule V remove hua aur source current placement Schedule II note karta hai; isliye Statement-I ko present context mein incorrect bataya gaya hai. [Q30]
- ▶ Indian Flying Fox frugivorous hai—fruits, nectar, flowers feed karta hai; blood-feeding statement bhi source ke mutabik incorrect hai. [Q30]

5. Amphibians, Frogs & Reptiles

- Amphibian ka literal sense “both kinds of life” hai—land aur water dono se associated; frogs cold-blooded/ectothermic hain aur moist skin se gas exchange bhi kar sakte hain. [Q34, Q35, Q36]
- Frog metamorphosis mein herbivorous tadpole ka long intestine adult carnivorous diet ke adaptation ke saath shorten hota hai. [Q37]
- Gharial/Gavialis ko source mainly Ganga River basin se associate karta hai. Otter ko Q39 ke source explanation mein given options ke context mein “atrocious/aggressive predatory” animal bataya gaya hai. [Q38, Q39]
- Hedgehog aur pangolin threat par body ko ball ki tarah curl karke vulnerable parts protect karte hain; marmot burrow mein hide karta hai. [Q40]

6. Squirrels, Dogs, Bears & Apes

- Indian squirrels mainly arboreal hain: tree nests/dreys banate hain, nuts/seeds ko ground mein store kar sakte hain, aur source unhe omnivorous batata hai. [Q41]
- Dingo bark nahi karta, wolf-like howl karta hai. Street-performing bear ke context mein source sloth bear answer deta hai. [Q42, Q43]
- Chimpanzee, parrot aur dolphin ko source language-use training ke examples mein rakhta hai; dog ko nahi. Human ko apes se separate karne ke liye source desire to learn aur ability to express ko highlight karta hai. [Q44, Q45]

- Langur ape nahi, Asian monkey hai aur tail hoti hai. Gibbon “lesser ape” aur source ke mutabik smallest man-like ape hai. Orangutan stick tools se tree/log holes se insects nikal sakta hai. Panda bear family Ursidae ka member hai. [Q46, Q47, Q48, Q49]

7. Heart Chambers, Gallbladder, Hooves & Wildlife Names

- Frog/amphibians aur most reptiles ka heart 3-chambered hota hai; fish 2-chambered, mammals/birds 4-chambered, aur crocodile 4-chambered diya gaya hai. [Q50, Q51]
- Jackal ke hooves nahi hote; source Canis aureus aur padded feet with nails mention karta hai. Nilgai ko source antelope/deer-type Bovidae family context mein rakhta hai. [Q53, Q54]
- Wildlife scientific-name match: Asiatic wild ass → Equus hemionus; Barasingha → Cervus duvaucelii; Chinkara → Gazella bennettii; Nilgai → Boselaphus tragocamelus. [Q55]
- Leopard ka scientific name Panthera pardus. Source lion breeding ko non-seasonal, cheetah ko non-roaring, aur leopard ko scent-marking big cat batata hai. [Q56, Q57]
- Marsupials naturally India mein nahi milte—source Statement-I ko true aur “sirf predator-free montane grasslands mein thrive karte hain” ko false batata hai. [Q58]

⚠ EXAM TRAP — Q52 has multiple valid animals in source

► Source explanation ke mutabik Camel, Giraffe aur Rat—teenon gallbladder ke bina listed hain. Isliye Q52 ko single-option fact ki tarah memorize na karein; source itself multiple options ko valid batata hai. [Q52]

8. Kiwi, Coral, Molluscs & Arthropods

- Match: wingless insect → Silverfish; flightless bird → Kiwi; limbless reptile → Snake; lungless creature → Fish. Kiwi New Zealand ka flightless bird hai. [Q59, Q60]
- Corals Cnidaria ke marine invertebrates hain aur reef-building calcium-carbonate skeleton ke karan source unhe organic origin ka batata hai. [Q61]
- Octopus cephalopod mollusc hai; source colossal squid ko largest invertebrate batata hai. [Q62, Q63]
- Crab, mite, spider, scorpion sab arthropods hain, lekin crab Malacostraca aur baaki three Arachnida mein diye gaye hain. White Leghorn fowl variety hai. [Q64, Q65]
- Ticks aur mites arachnids hain; termites “white ants” ke naam se bhi known hain; glow-worm luminous insect hai. [Q66, Q67, Q68]
- Parasite ke andar parasite ko hyperparasitism/hyperparasite phenomenon kaha gaya hai. Female Anopheles resting position mein body substrate se angle banati hai, while Culex more parallel rest karta hai. [Q69, Q70]
- Spider insect nahi, arachnid hai—8 legs; insects ke 6 legs. Spider silk ko source Gossamer silk bhi kehta hai. [Q71, Q72, Q73, Q74]

9. Silk, Lac & Insect Physiology

- Commercial silk silkworm ke pupa/cocoon stage mein milta hai; Bombyx mori mulberry leaves par feed karta hai. [Q75, Q76]
- Lac lac insects ki resinous secretion hai. Silk, lac aur honey insects se milte hain, pearl molluscs se—so pearl insect product nahi. [Q77, Q78]
- Cockroach ka blood white/colourless diya gaya hai because haemoglobin absent; oxygen transport tracheal tubes se hota hai. Earthworm mein red blood cells absent hain but haemoglobin plasma mein dissolved hota hai. [Q79, Q80]
- Most insects tracheal system se respire karte hain. Cicada, frogopper aur pond skater source ke mutabik insects hain. [Q81, Q82]

10. Parasites, Feeding Types & Nocturnal Animals

- Tapeworm hermaphrodite; roundworm mein separate sexes; filaria nematode se; Guinea worm bhi nematode hai, annelid nahi. [Q83]
- Detritivores mein earthworm, millipede aur woodlice included; jellyfish/seahorse nahi. Filter-feeder examples mein clams, oysters, krill, sponges, baleen whales aur kuch fish diye gaye hain. [Q84, Q85]
- Parasitoid species source ke mutabik carabid beetles, flies aur wasps mein milti hain; centipedes predators aur termites detritivores diye gaye hain. [Q86]
- Mosquito, bat, owl aur kiwi nocturnal examples hain. Given fauna mein source Malabar civet ko strictly nocturnal answer leta hai; lion-tailed macaque diurnal aur sambar mainly crepuscular/nocturnal activity wala bataya hai. [Q87, Q88]
- Ant ka brain body-size proportion ke context mein source sabse large batata hai; ant bite mein formic acid inject hota hai, aur kuch ant species fungi cultivate karti hain. [Q89, Q90, Q91]
- Blood sucking female mosquitoes karti hain; males flower/fruit juices par feed karte hain. Source female mosquitoes ko generally larger/bigger-winged batata hai. [Q92]

11. Desert Animals & Snake Biology

- Camel ke context mein source intestine mein water storage mention karta hai. Kangaroo rat ko longest without drinking water aur highly concentrated/"solid" urine excretion se associate karta hai. [Q93, Q94, Q95]
- Female spider mating ke around male ko kill/eat kar sakti hai—source ise spider cannibalism ke example ke roop mein deta hai. [Q96]
- King cobra nest banata hai eggs lay/guard karne ke liye; source ise only snake that builds a nest batata hai. King cobra ka diet mainly other snakes hai. [Q97, Q98, Q100]
- Cobra venom mainly neurotoxic diya gaya hai. Python non-venomous snake hai; snake poison glands vertebrate salivary glands ke homologous aur fangs modified maxillary teeth diye gaye hain. [Q99, Q101, Q102, Q103]
- Source ke mutabik butterflies, fish aur frogs—teeno groups mein poisonous species mil sakti hain. Kuch turtles/fish/marine mammals herbivorous aur kuch snakes viviparous ho sakte hain. [Q104, Q105]
- Dog ko source herbivore nahi maanta aur available evidence ko natural carnivory ke support mein frame karta hai. [Q106]

12. Insectivorous Plants & Pitcher Plant

- Rafflesia arnoldii source ke mutabik world ka largest flower hai; plant leaves/roots/stem ke bina described hai. [Q107]
- Insectivorous plants nitrogen-deficient soil mein grow karte hain aur insects se nitrogen requirement fulfill karte hain. Pitcher plant/Nepenthes insectivorous plant hai. [Q108, Q109, Q112]
- Pitcher modified leaf hota hai jo pitfall trap ki tarah work karta hai. Nepenthes photosynthetic autotroph hai, parasite nahi. [Q110, Q111]
- Nepenthes khasiana India ka rare/endangered pitcher plant hai aur source ise Garo, Khasi, Jaintia hills of Meghalaya se associate karta hai. [Q113]

13. Underground Stems, Roots & Storage Organs

- Turmeric Curcuma longa ka underground modified stem—rhizome—hai; turmeric powder aur edible part bhi isi rhizome se aata hai. [Q114, Q115, Q116, Q117]
- Sweet potato modified root hai; source potato, Bandda aur Jimikand ko modified stems ke group mein rakhta hai. Sugarcane, potato, ginger ke storage organs stem hain; sweet potato ka root. [Q118, Q119]

- Ginger modified stem hai; nodes, internodes aur scaly leaves uski stem nature indicate karte hain. Turnip, carrot aur sweet potato modified roots diye gaye hain. [Q120, Q121]
- Match: Rhizome → Ginger; Tuber → Potato; Bulb → Garlic; Corm → Jimikand/Suran. [Q122]
- Potato *Solanum tuberosum* ka underground modified stem/tuber hai. [Q123, Q124]

14. Pea Family, Clove & Saffron

- Pea plant dicot herbaceous annual hai aur Fabaceae/Leguminosae family mein aata hai. Groundnut, horse-gram aur soybean—teeno pea family/Fabaceae legumes hain. [Q125, Q126]
- Clove *Syzygium aromaticum* ka aromatic dried flower bud hai; MCQ wording mein dried flower/flower bud dono form source mein use hue hain. [Q127, Q128, Q129, Q130, Q131, Q132]
- Saffron *Crocus sativus* flower ke stigma aur styles se milta hai, seed se nahi. Source Q134 mein ise gynoecium/female reproductive “seed-forming parts” ke context mein describe karta hai. [Q133, Q134]

15. Edible Plant Parts, Fruits & Plant Chemicals

- Potato ka edible part tuber/modified stem hai; onion ka edible portion fleshy modified leaves aur coconut ka cellular endosperm diya gaya hai. [Q135, Q136]
- Ladies finger/okra botanically fruit hai; potato stem aur radish/sweet potato roots hain. Source apple ko false fruit ke example ke roop mein deta hai; lychee ko broad structural category “drupe” context mein place karta hai. [Q137, Q138, Q139]
- Garlic ki characteristic smell sulphur compounds (Allicin/Allyl methyl sulphide) se; onion cutting par sulfenic acid lacrimal glands irritate karta hai; chilli pungency capsaicin se hoti hai. [Q140, Q141, Q142, Q143]
- Fruit sweetness ka source answer fructose hai. Diatoms drowning diagnosis mein forensic significance rakhte hain. Mulberry ka multiple fruit type sorosis diya gaya hai. [Q144, Q145, Q146]

16. Plant Fibres, Triticale & Dry-land Adaptations

- Sunn, hemp aur jute bast/stem fibres hain; cotton seed-hair fibre hai, stem product nahi. [Q147]
- Triticale wheat × rye cross se developed first man-made cereal crop diya gaya hai; source ise nature mein naturally found nahi batata. [Q148, Q149]
- Bare rock/soil-less surface par lichens aur mosses survival ke likely pioneers diye gaye hain; ferns generally soil aur mushrooms decaying organic matter/soil se associated hain. [Q150]
- Desert plants ki long roots deep water search ke adaptation ke roop mein di gayi hain; high soil temperature ko cause nahi maana gaya. Phreatophytes deep groundwater tap karne wali deep-rooted plants hain. [Q151, Q152, Q153]
- Xerophytes dry climate adapted plants hain; succulents thick fleshy tissues mein water store karte hain. [Q154, Q155]

17. Plant Habitat Types — Epiphyte, Halophyte, Hydrophyte & Lithophyte

- Epiphytes host plant par grow karke mechanical support lete hain, food/water directly host se nahi; source orchids ko example deta hai. [Q156]
- High-salinity soil/water plants halophytes kehlate hain. Hydrophytes aquatic plants hote hain; lithophytes rocks par grow karte hain. [Q157, Q158, Q159, Q160]
- *Nepenthes* hydrophyte nahi—source ise tropical carnivorous pitcher plant batata hai. [Q161]
- Q162 source ke mutabik weeping jelly excess acid se; sugarcane ke liye *Acetobacter diazotrophicus* *Rhizobium* se more suitable; salt-water plants halophytes; dormancy breaking mein gibberellins/cytokinin use ho sakte hain. [Q162]
- Land par living plants/animals terrestrial organisms kehlate hain. [Q163]

18. Ecology — Producers, Decomposition & Trophic Levels

- Producers/autotrophs biotic aur abiotic components ke intermediaries ke roop mein source mein described hain—sunlight, water, CO₂ jaise abiotic resources ko organic food mein convert karte hain. [Q164]
- Microorganisms ke through organic matter breakdown decomposition hai; bacteria/fungi decomposers complex organics ko simpler substances mein convert karte hain. [Q165]
- Autotroph = producer; saprotroph dead organic matter se nutrients; herbivores primary consumers hote hain, secondary consumers nahi. [Q166]
- Trophic level-I mein autotrophs/green plants aate hain. Ocean primary producers ke given set mein cyanobacteria aur diatoms hain; copepods/foraminifera consumers diye gaye hain. [Q167, Q168]

19. Fish Survival, Respiration & Plant Pigments

- Frozen lakes mein upper surface freeze hone ke baad below liquid water remain karta hai; source anomalous expansion / 4°C density behavior se fishes ke survival ko explain karta hai. [Q169]
- Fish gills dissolved oxygen absorb karte hain; water se bahar gills atmospheric oxygen efficiently absorb nahi kar pate, isliye fish die kar sakti hai. [Q170, Q171]
- Apple ka red colour anthocyanin se; tomato ka red colour lycopene se. [Q172, Q173, Q174, Q175]
- Papaya yellow colour source caricaxanthin pigment se associate karta hai; papaya carotene-rich aur papain enzyme mention kiya gaya hai. [Q176]
- BSI scientists ke source note ke mutabik ~11 m tall, orange-pulp banana species *Musa indandamanensis* Little Andaman Islands ke Krishna-Nalah forest mein discover hui. [Q177]

20. Species, Fungi & Mushrooms

- Chinese, Americans, Indians aur Black Africans source ke mutabik same human species *Homo sapiens* ke members hain. [Q178]
- Fungi chlorophyll lack karte hain aur separate kingdom mein classified hain. Yeast single-celled fungus hai; yeast aur mushrooms Kingdom Fungi ke members hain. [Q179, Q180, Q181]
- Mushroom fungus ka fleshy spore-bearing fruiting body hai; edible cooking mushrooms bhi fungi hain. [Q182, Q183]
- Gucchi/morels fungus hain aur Himalayan forest regions mein wild grow karte hain; source commercial cultivation statement ko incorrect batata hai. [Q184]

21. Symbiosis — Fig, Mycorrhiza, Lichen & Rhizobium

- Fig tree species aur specific fig-wasp pollinator ke beech one-to-one coevolved relationship source ka classic coevolution example hai. [Q185]
- Cnidarians, fungi aur protozoa—teeno symbiotic relationships establish kar sakte hain; source corals—dinoflagellates, fungi—plants/algae, aur protozoan mutualisms examples deta hai. [Q186]
- Mycorrhiza fungi aur higher-plant roots ka beneficial symbiotic association hai: fungus water/nutrient absorption help karta hai, plant carbohydrates provide karta hai. [Q187, Q188]
- Lichen fungus + alga/cyanobacterium ka symbiotic association hai; interaction mutualism hai. Algal partner photosynthesis se organic compounds deta hai, fungal partner moisture/shelter. [Q189, Q190, Q191, Q192]
- Lichens atmospheric pollution ke sensitive bio-indicators hain aur polluted areas mein grow nahi karte. Leguminous root nodules ke nitrogen-fixing bacteria (e.g., *Rhizobium*) source ke mutabik symbiotic hain. [Q193, Q194]